Hackathon Title

NETWORK ATTACK DETECTION USING KALI LINUX

Hackathon Report submitted
for the partial fulfilment of grade for

the subject

COMPUTER NETWORKS (24CS2202)

Submitted

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ABSTRACT

This project presents the detection and analysis of network attacks using Kali Linux in a virtualized environment. With the rapid growth of computer networks, cyber attacks such as port scanning, denial-of-service, and unauthorized access have increased significantly. The project focuses on simulating real-world attacks from an attacker system using Kali Linux and monitoring them on a victim system using Ubuntu Linux. Network monitoring tools like Snort (Intrusion Detection System) and Wireshark are used to detect suspicious activities and analyze network packets. The system generates alerts for malicious traffic and helps in understanding attack patterns. This project provides hands-on exposure to ethical hacking, intrusion detection, and network security concepts, making it useful for learning practical cybersecurity techniques.

Problem statement

Traditional network security mechanisms such as firewalls are not sufficient to detect all types of cyber attacks. Existing systems often fail to identify internal threats, port scanning activities, and denial-of-service attacks in real time. Many networks lack proper monitoring tools to analyze traffic and generate alerts for suspicious behavior. This creates vulnerabilities that attackers can exploit, leading to data breaches and service disruptions. Therefore, there is a need for an effective intrusion detection system to monitor network traffic and detect malicious activities efficiently.

Objectives

1. To simulate common network attacks using Kali Linux in a controlled environment.
2. To detect and analyze malicious network traffic using Snort and Wireshark.
3. To understand the working of intrusion detection systems for improving network security.

Methodology

1. Install Kali Linux (attacker) and Ubuntu Linux (victim) using VirtualBox.
2. Configure both virtual machines on the same NAT network.
3. Perform network scanning and attack simulations using Nmap and Hping3.
4. Monitor network traffic on Ubuntu using Snort and Wireshark.
5. Analyze alerts and captured packets to identify attack patterns.

Conclusion

1. The project successfully demonstrates detection and analysis of network attacks.
2. Intrusion detection systems generate alerts for malicious activities.
3. The project improves understanding of network security mechanisms.

Software and hardware requirements

Software Requirements

1. Oracle VirtualBox
2. Kali Linux OS
3. Ubuntu Linux OS
4. Snort IDS
5. Wireshark
6. Nmap
7. Hping3

Hardware Requirements

1. Laptop/Desktop Computer
2. Minimum 8 GB RAM (Recommended)
3. Intel/AMD Processor with Virtualization Support
4. Minimum 50 GB Free Disk Space
5. Stable Internet Connection